

COMPLETE HVAC PLANT FLOW DIAGRAM – IIT INDORE

COOLING TOWER SYSTEM (OPEN CYCLE)



Cold Water
(25°C)

Hot Water
(30°C)



FILL / PACKING
(Heat Transfer Media)

WATER
MAKEUP TANK



STRAINER



CONDENSER WATER PUMP
(OPEN CYCLE)



RETURN CHILLED WATER
(9 – 11 °C)



(Access for maintenance) (Supply & Return)

HOSTELS



AHU / 450 TR

LHC (LECTURE HALL COMPLEX)



AHU / 450 TR

PODS



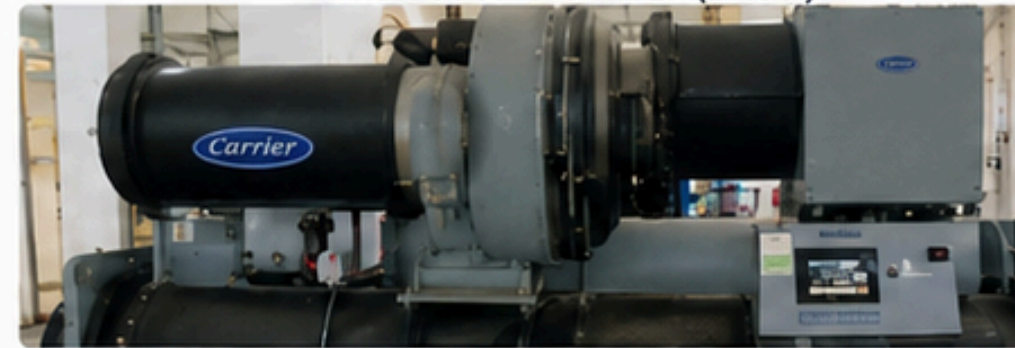
AHU / 450 TR

OTHER BUILDINGS



AHU / 450 TR

WATER COOLED CHILLER (2 NOS.)



2 NOS. CHILLER (2 WORKING) 650 TR EACH

1. CONDENSER



Refrigerant rejects
heat to water

2. COMPRESSOR



Increases pressure
& temperature

3. EXPANSION VALVE



Reduces pressure
& temperature

4. EVAPORATOR



Refrigerant absorbs
heat from water

CHILLER CONTROL PANEL



TYPICAL OPERATION (EXAMPLE)

CHILLED WATER (EVAPORATOR)

Entering : 10.2 °C
Leaving : 6.9 °C
Approach : 0.5 °C

CONDENSER WATER

Entering : 30.5 °C
Leaving : 33.7 °C
Range : 3.2 °C

EXPANSION
TANK



AIR
SEPERATOR
TANK



CHILLED WATER PUMP SYSTEM (CLOSED LOOP)

PRIMARY PUMP
(Return from Buildings)

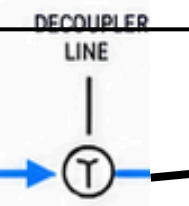


Inlet Pressure: 3 KG
Outlet Pressure: 5 KG

SECONDARY PUMP
(Supply to Buildings)



Inlet Pressure: 4 KG
Outlet Pressure: 8.5 – 9 KG



Chilled Water
(6 – 7 °C)

DECOUPLER
LINE



MAIN HEADER



UNDERGROUND CHILLED WATER DISTRIBUTION NETWORK (CLOSED LOOP)

CHILLED WATER CYCLE (CLOSED LOOP)

Heated water return from buildings is sucked by the Primary Pump. Before suction, water passes through strainers and pressure maintenance pump maintains line pressure at 4 KG. This return water is sent to the chiller condenser, cooled and then sent to supply via Secondary Pump and Decoupler Line (with desired load).

REFRIGERATION CYCLE (INSIDE CHILLER)

Refrigerant cycle takes place inside the chiller. The chiller removes heat from the condenser water (closed loop) and transfers it to the condenser water (open loop).

CONDENSER WATER CYCLE (OPEN LOOP)

Heated condenser water is sent to the Cooling Tower and gets cooled there (open cycle). The cooled water is then sucked again by the Condenser Pump and sent back to the chiller condenser.

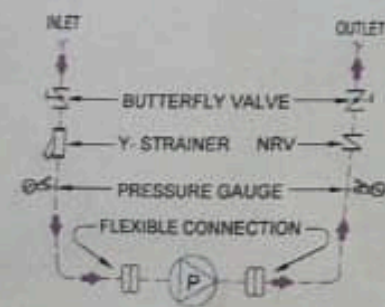
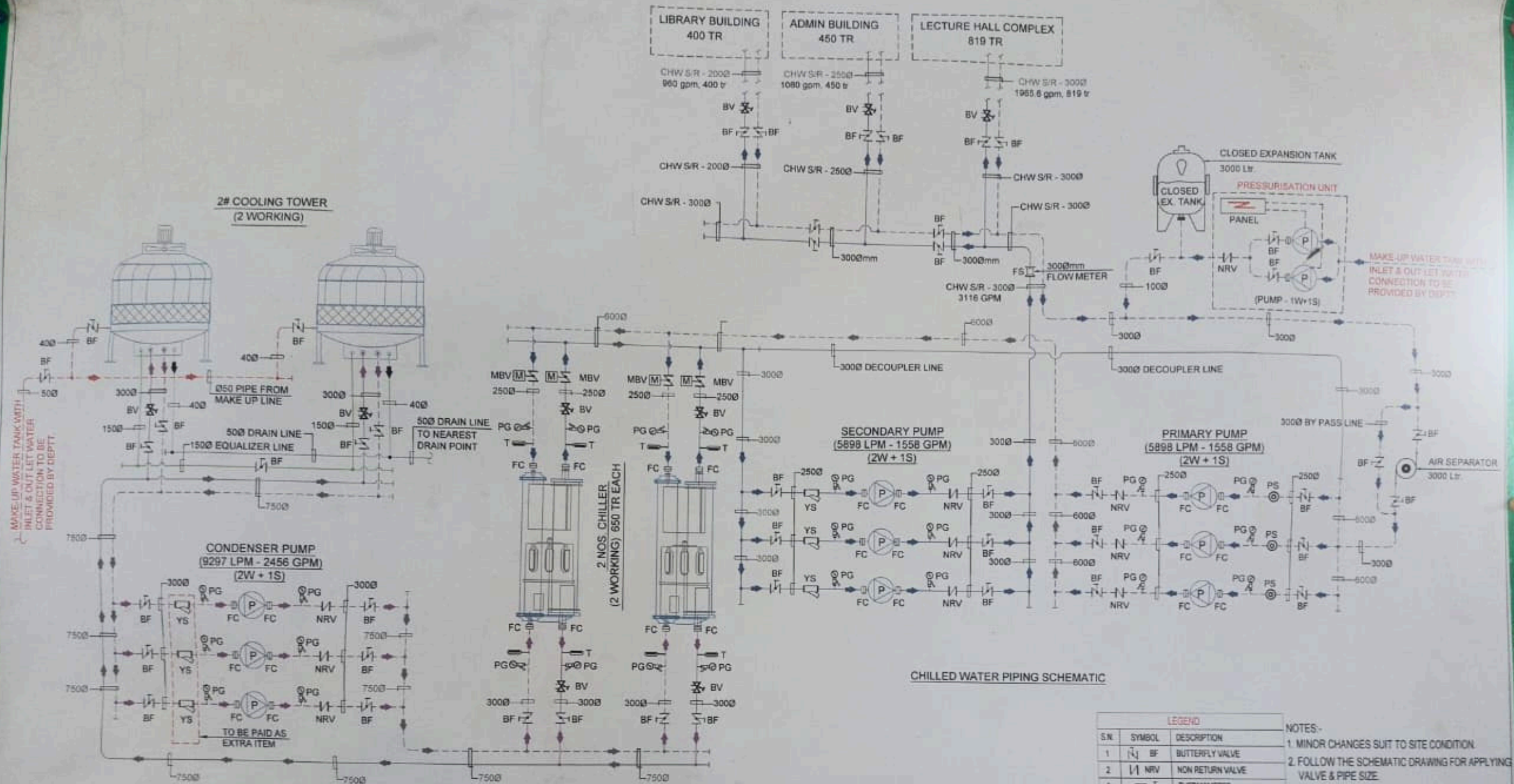
Property	R134a	R410A	R32	R1234yf
Chemical Formula	CH ₂ FCF ₃	R32/R125 Blend	CH ₂ F ₂	C ₃ H ₂ F ₄
GWP (100 yr)	~1430	~2088	~675	<1
Boiling Point (1 atm)	-26.1°C	-51.6°C	-51.7°C	-29.5°C
Critical Temperature	101°C	72.5°C	78.1°C	94.7°C
Volumetric Cooling Capacity	Medium	High	Very High	Medium
Safety Class	A1	A1	A2L	A2L
Typical Applications	Chillers, HVAC	Split AC, VRF	Inverter ACs	Automotive AC

UNDERGROUND RETURN
AND SUPPLY LINES

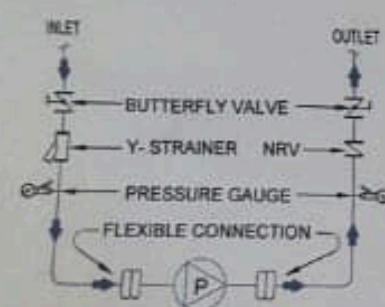


CURRENT COP

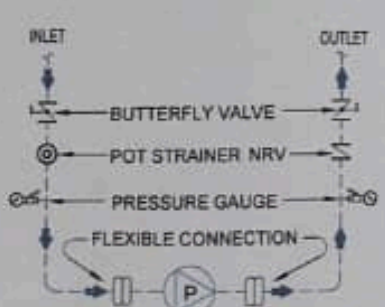
4 -5



TYPICAL CONDENSER PUMP
CONNECTION WITH VALVE



TYPICAL SECONDARY PUMPS
CONNECTION WITH VALVE



TYPICAL PRIMARY PUMPS
CONNECTION WITH VALVE

CHILLED WATER PIPING SCHEMATIC

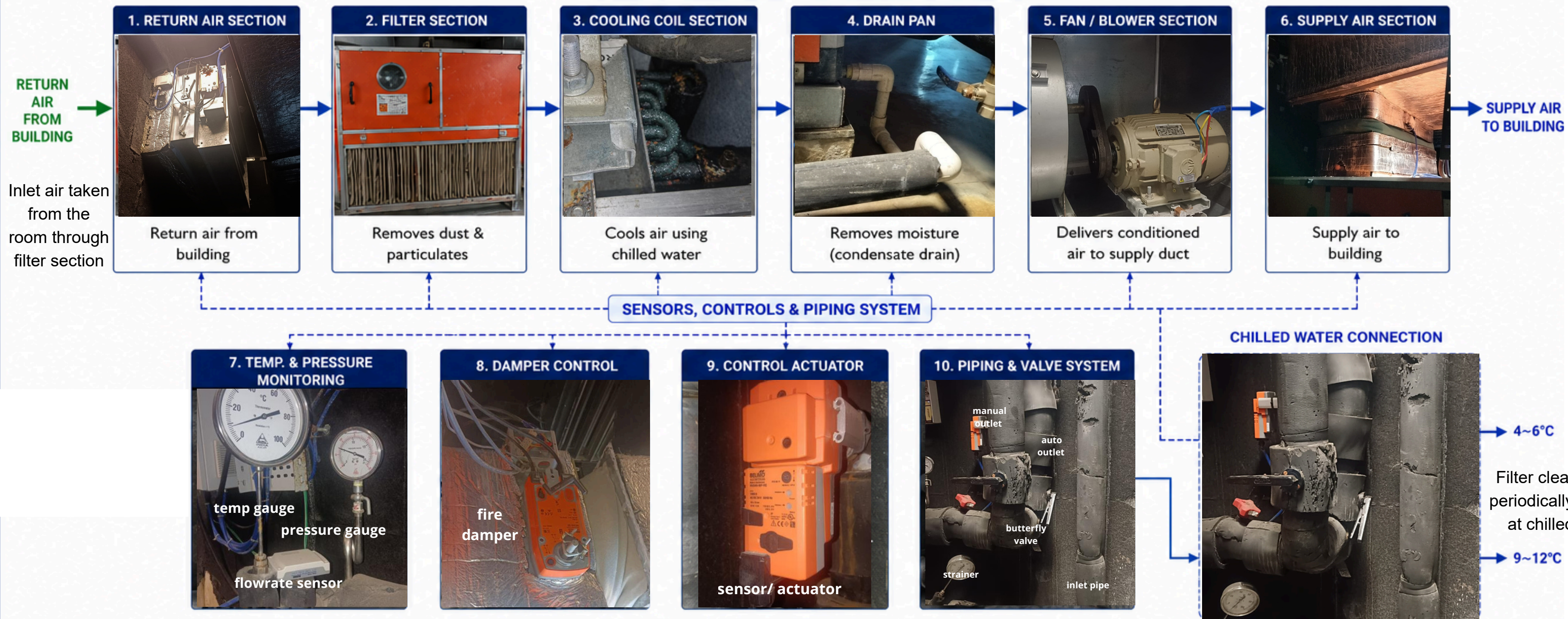
LEGEND	
S.N.	SYMBOL DESCRIPTION
1	BF BUTTERFLY VALVE
2	NRV NON RETURN VALVE
3	T THERMOMETER
4	PG PRESSURE GAUGE
5	YS Y-STRAINER
6	BV BALANCING VALVE
7	AS AIR SEPARATOR
8	PS POT STRAINER
9	PUMP
10	MBV MOTORIZED BUTTERFLY VALVE
11	CHILLED WATER SUPPLY PIPE
12	CHILLED WATER RETURN PIPE
13	CONDENSER WATER S PIPE
14	CONDENSER WATER R PIPE
15	EXISTING PIPING
16	FC RUBBER FLEXIBLE ELLOWS

- NOTES:-
- MINOR CHANGES SUIT TO SITE CONDITION.
 - FOLLOW THE SCHEMATIC DRAWING FOR APPLYING VALVE & PIPE SIZE.

PROJECT		IIT INDORE	
REV.	DATE	DESCRIPTION	BY
02	18.8.2020	REVISED AS PER E/E	M.S. S.K.
01	03.3.2020	REVISED AS PER E/E	M.S. S.K.
DRG. TITLE:-			
CHILLED WATER PIPING SCHEMATIC			
Drawn	Checked	Approved	Scale
1:1	1:1	1:1	1:1
airf engineers (p) ltd.			
B-23, INFOCITY SECTOR-34 GURGAON-122001 HARYANA			

AHU (AIR HANDLING UNIT) – WORK FLOW DIAGRAM

AHU AIR FLOW SEQUENCE (1 TO 6)



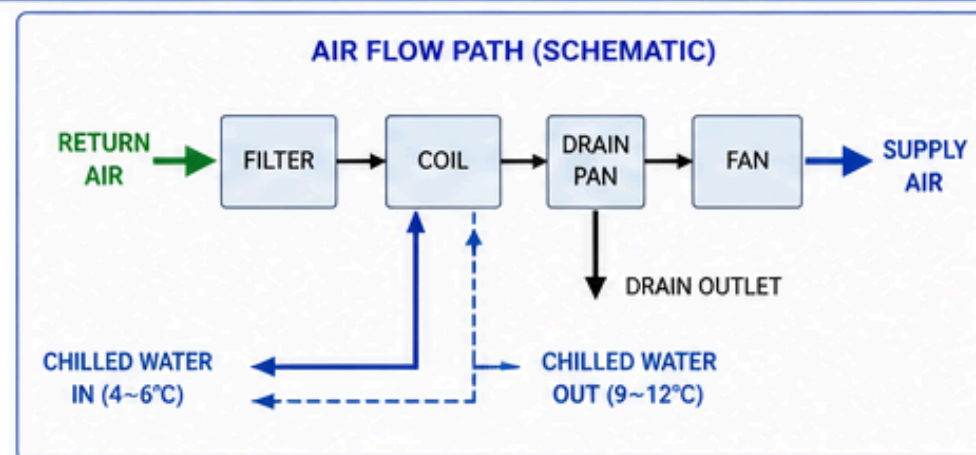
OPERATING TEMPERATURE (TYPICAL)

RETURN AIR : 24~26 °C

SUPPLY AIR : 14~16 °C

CHILLED WATER IN : 4~6 °C

CHILLED WATER OUT : 9~12 °C



ZECO AIRCON LIMITED			
AN ISO 9001:2015 COMPANY			
PH. : 91 1276 244650-53, FAX : 91 1276 244568			
Website : www.zecoaircon.com			
MODEL NO.	ZDSV-60	FAN R.P.M.	1176
SERIAL NO.	18279132-8	MOTOR KW/POLE/PH	3.7/4/3
AIR QTY. (CFM)	6000	COIL	CHW
ST. PRESSURE (MMWG)	50	FILTER	EU-6
FAN MODEL	FDA-12X12-S2	YEAR OF MANUFACTURING	02/2019
FAN PULLEY DIA(MM)	132X2SPA	MOTOR PULLEY DIA(MM)	109X2SPA
TAG NO.	ZEC/AC/2025/060		